

LAST TIME:

$$E(g(x)) = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

" $E(Y)$

FUNCTION PDF



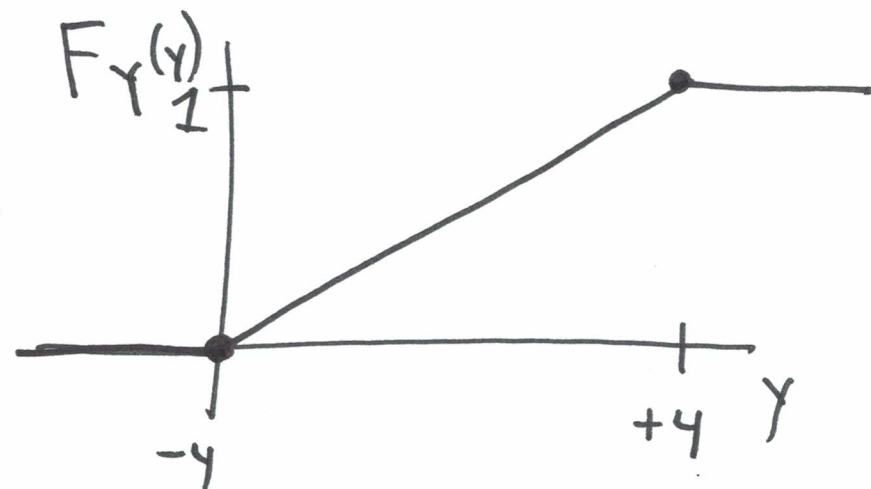
$Y = g(x)$ IS ALSO A RANDOM VARIABLE.

WHAT IS THE FULL PDF OF $Y = g(x)$?

EASIEST WAY IS TO COMPUTE THE
CDF OF Y AND TAKE DERIVATIVE.

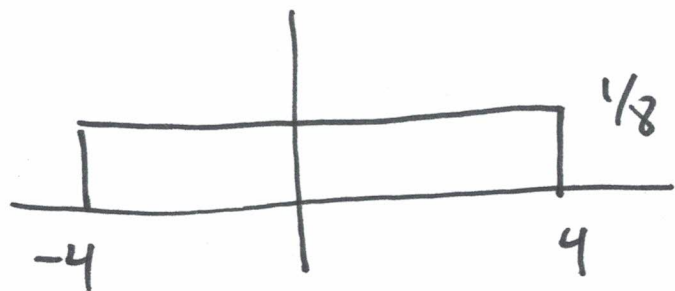
$$= \int_0^{(y+4)^{1/3}} \frac{3}{8} x^2 dx$$

$$= \left. \frac{1}{8} x^3 \right]_{x=0}^{x=(y+4)^{1/3}}$$



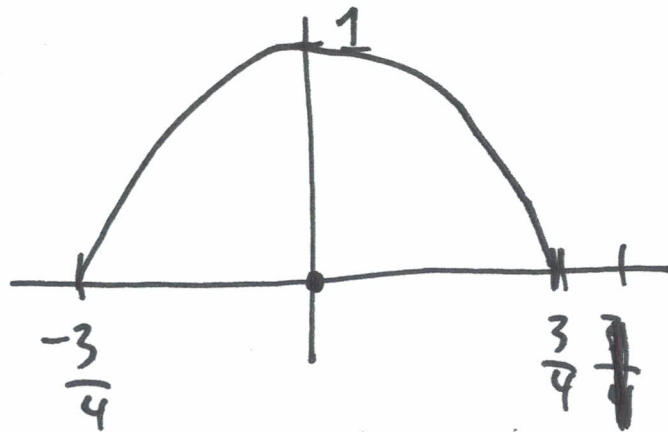
$$= \begin{cases} \frac{1}{8}(y+4) & \text{For } y \in [-4, 4] \\ 0 & \text{For } y < -4 \\ 1 & \text{For } y > 4 \end{cases}$$

$$\text{PDF of } Y: \frac{d}{dy} F_Y(y) = \begin{cases} \frac{1}{8} & y \in [-4, 4] \\ 0 & \text{Else} \end{cases}$$



WE NEED TO LOOK CAREFULLY AT THE
RANGES OF X AND Y IN CASE
IT'S NOT 1-TO-1.

EX $f_X(x) = 1 - \frac{16}{9}x^2 \quad x \in \left[-\frac{3}{4}, \frac{3}{4}\right]$



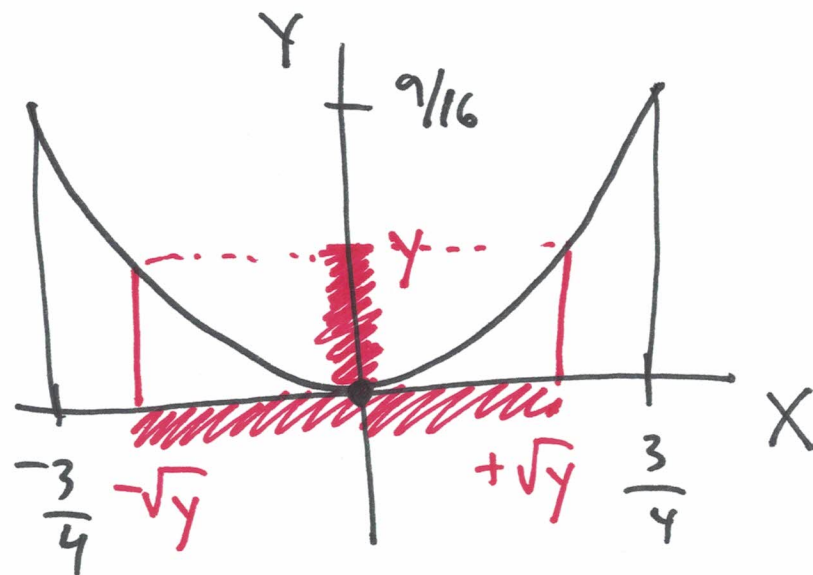
LET $Y = X^2$. WHAT IS PDF OF Y ?

Y RANGES FROM 0 TO $\frac{9}{16}$.

CDF of Y : $F_Y(y) = P(Y \leq y)$

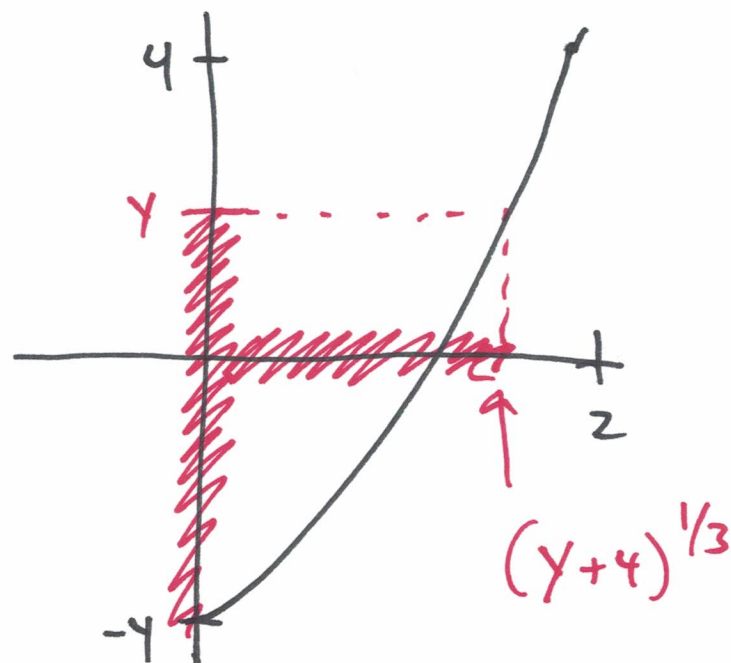
$$P(X^2 \leq y)$$

$$P(X \in [-\sqrt{y}, +\sqrt{y}])$$



VS THE FIRST EXAMPLE

$$\int_{-\sqrt{y}}^{\sqrt{y}} 1 - \frac{16}{9}x^2 dx$$



$$x - \frac{16}{27} x^3 \Bigg|_{x=-\sqrt{y}}^{x=\sqrt{y}} = 2 \left(\sqrt{y} - \frac{16}{27} \sqrt{y}^3 \right)$$

CDF.

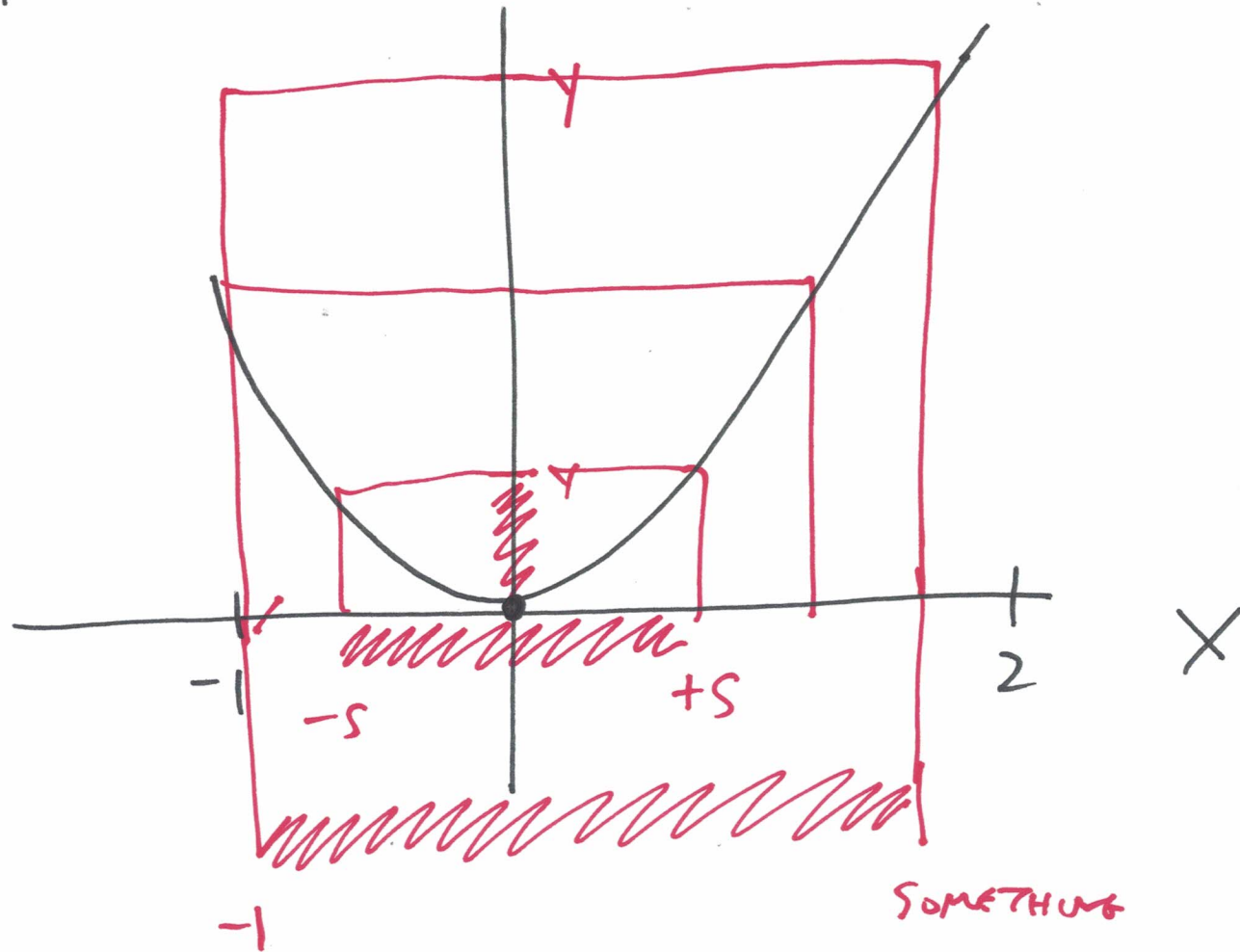
~~4/2~~ PDF = $\frac{d}{dy} F_Y(y)$

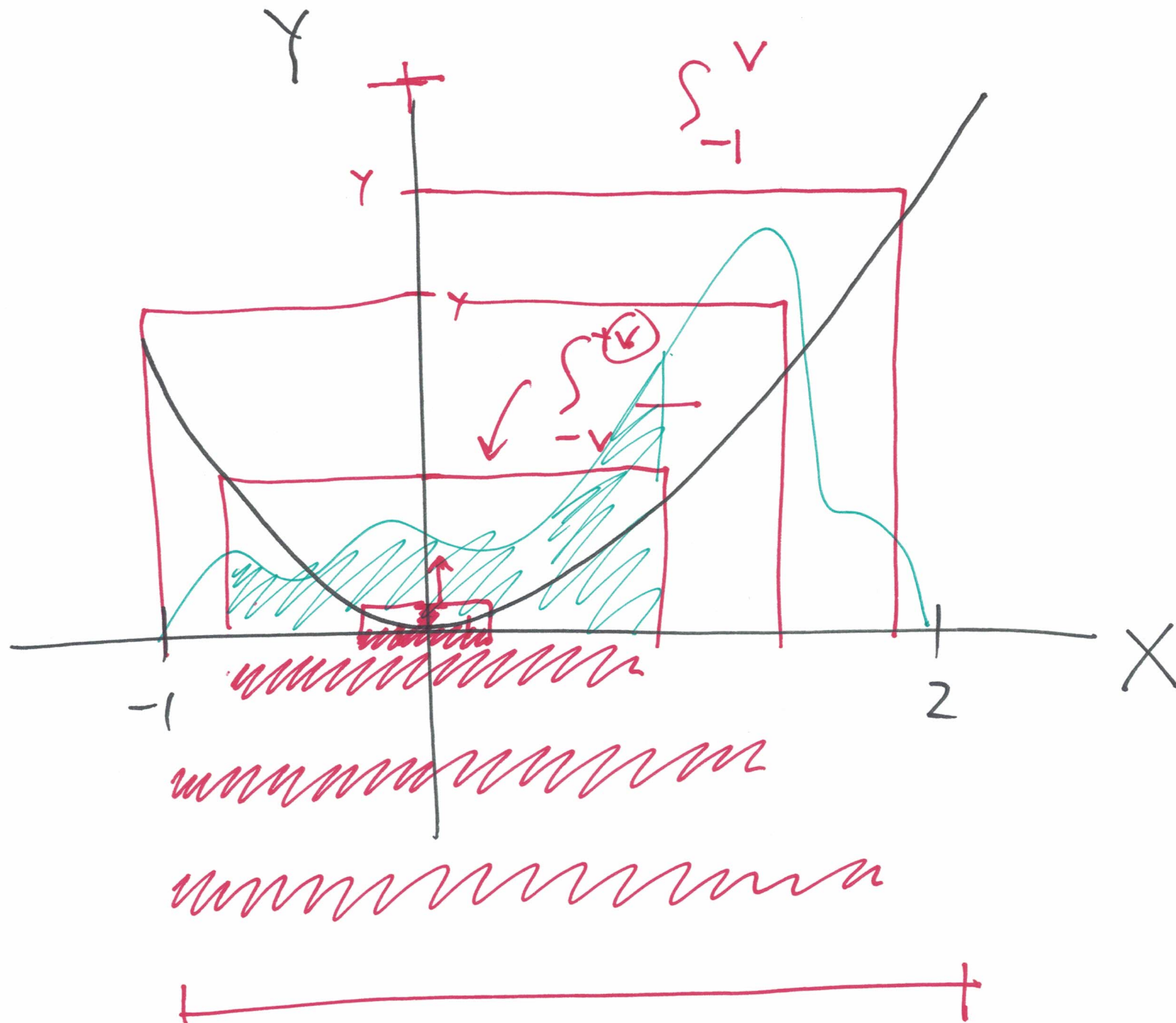
$$\frac{d}{dy} 2 \left(y^{\frac{1}{2}} - \frac{16}{27} y^{\frac{3}{2}} \right)$$

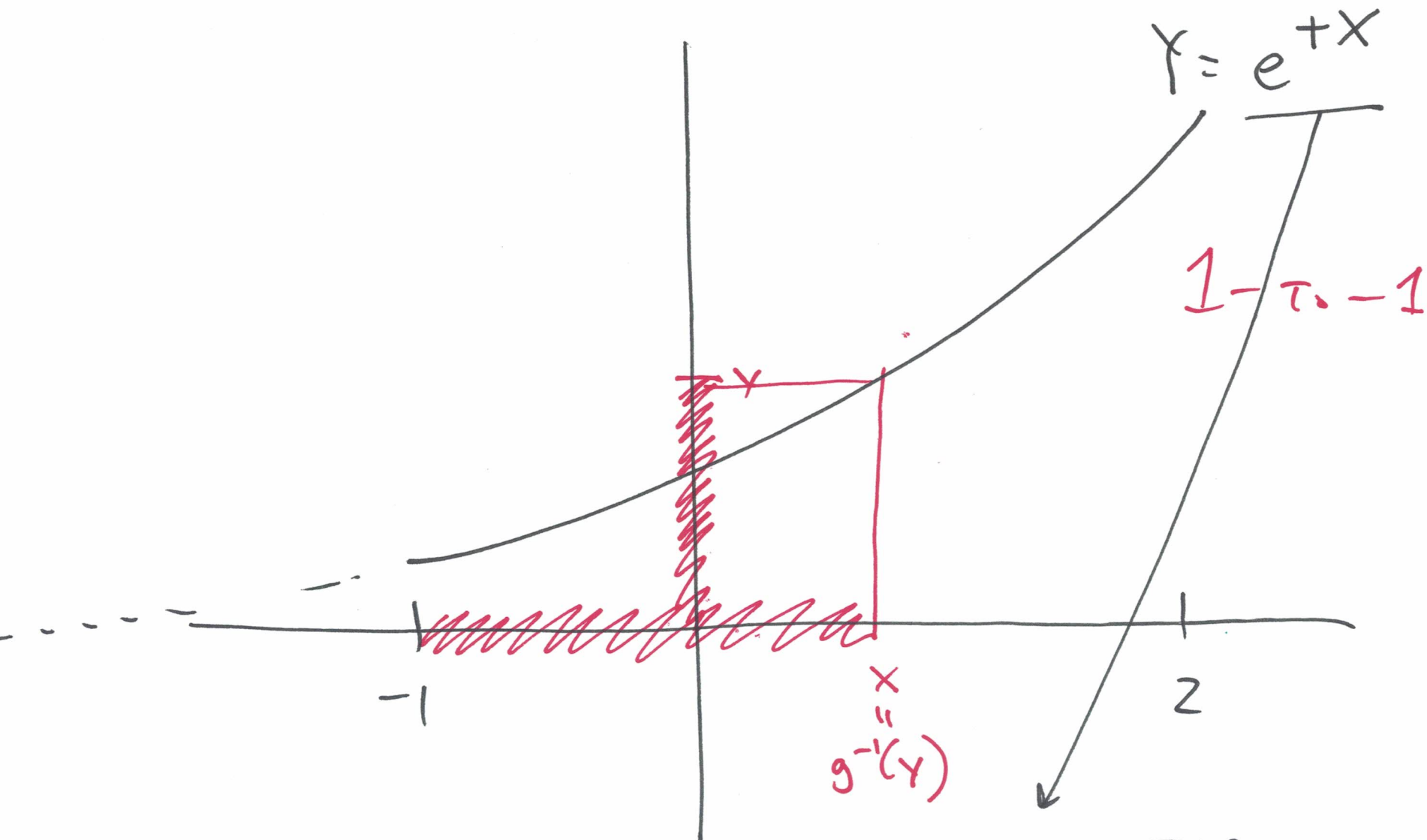
$$= \begin{cases} y^{-\frac{1}{2}} - \frac{16}{9} y^{\frac{1}{2}} & y \in [0, \frac{9}{16}] \\ 0 & \text{OTHERWISE.} \end{cases}$$

HW 5

PREVIEW:







WATCH THIS
FUNCTION.

CAN THE SAME ~~OF~~
Y RESULT FROM
MULTIPLE X'S?

IF EVERYTHING IS 1-TO-1, THERE
IS A DIRECT FORMULA:

$$\begin{aligned} F_Y(y) &= P(Y \leq y) = P(g(x) \leq y) \\ &= P(X \leq \underbrace{g^{-1}(y)}_{\text{UNIQUE}}) = F_X(g^{-1}(y)) \end{aligned}$$

$$\begin{aligned} f_Y(y) &= \frac{d}{dy} F_X(g^{-1}(y)) \\ &= f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y) \end{aligned}$$

LIKE IN THE FIRST EXAMPLE,

$$y = g(x) = x^3 - 4 \quad f_x(x) = \frac{3}{8} x^2$$

$$g^{-1}(y) = (y+4)^{\frac{1}{3}}$$

$$\frac{3}{8} \cancel{(y+4)^{\frac{2}{3}}} \cdot \frac{1}{3} \cancel{(y+4)^{\frac{2}{3}}} = \frac{1}{8}$$

SAME THING, DIFFERENT WAY.

IN GENERAL

(*) EXTRA INFO

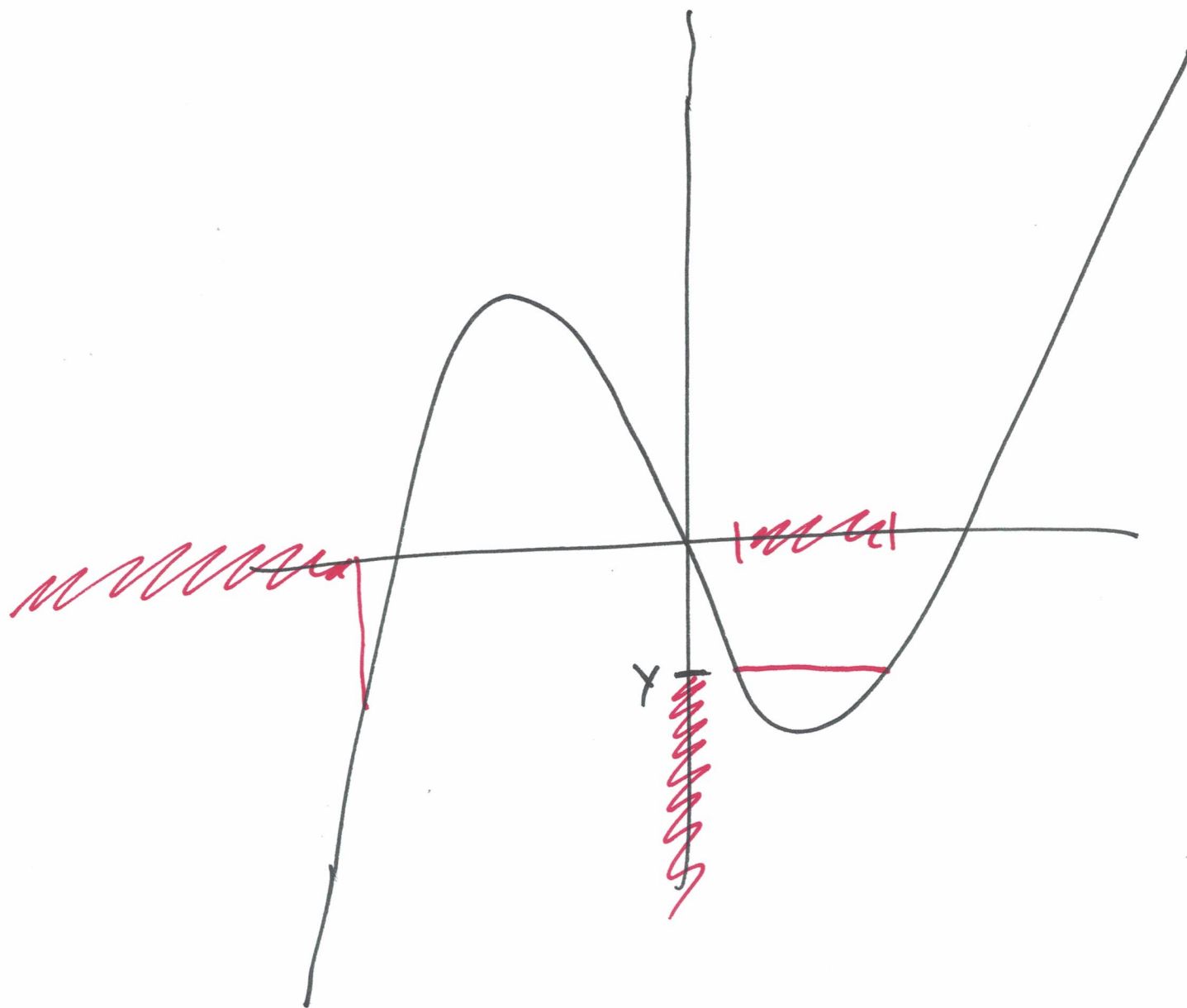
IF $Y = g(X)$ NOT ONE TO ONE,

AND LET $\{x_1, \dots, x_k\}$ THAT ALL MAP TO y

$$g(x_i) = y$$

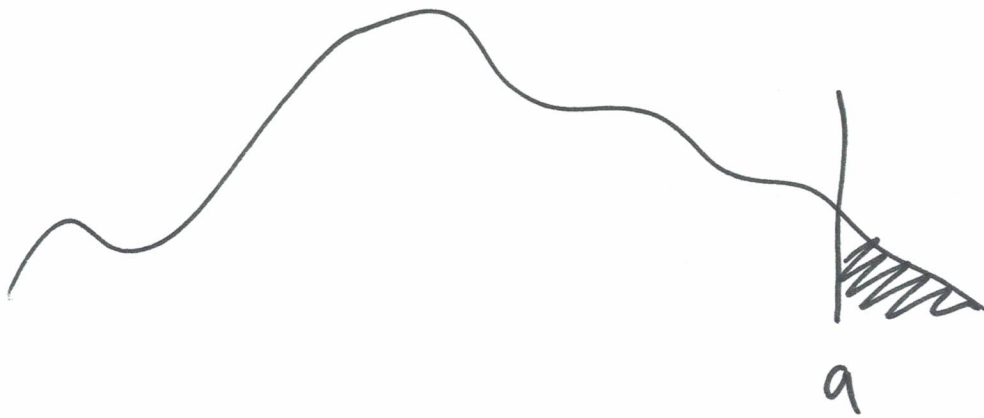
THEN

$$f_Y(y) = \sum_{i=1}^k f_X(x_i) \cdot \frac{1}{|g'(x_i)|}$$



NOTES ON INEQUALITIES:

USED TO BOUND WORST-CASE SCENARIOS
ON UNLIKELY EVENTS.



$$P(X \geq a)$$

SETUP: X IS NON-NEGATIVE.

MARKOV INEQUALITY:
$$P(X \geq a) \leq \frac{E(X)}{a}$$

UPPER BOUND ON THE TAIL.

Ex X TURTLE $E(X) = 300$

$$P(X > \underline{350}) \leq \frac{300}{350} = \frac{6}{7} = 0.857$$

NOW SUPPOSE WE ALSO KNOW $\text{Var}(X) = \sigma^2$
 $= 1728$

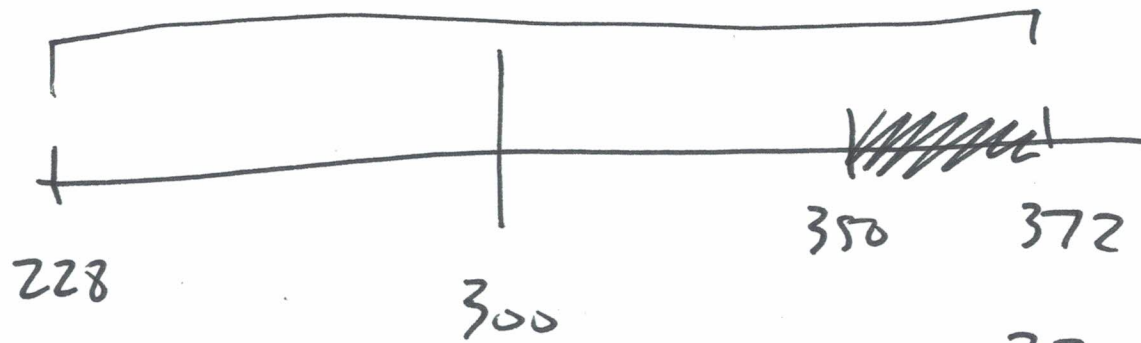
WE CAN APPLY CHEBYSHEV'S INEQUALITY:

$$P(|X - \underbrace{\mu}_{300}| \geq \underbrace{b}_{50}) \leq \left(\frac{\underbrace{\sigma}_{\sqrt{1728}}}{\underbrace{b}_{50}} \right)^2$$

$$P(|X - 300| \geq 50) \leq \frac{1728}{2500} = 0.6912.$$

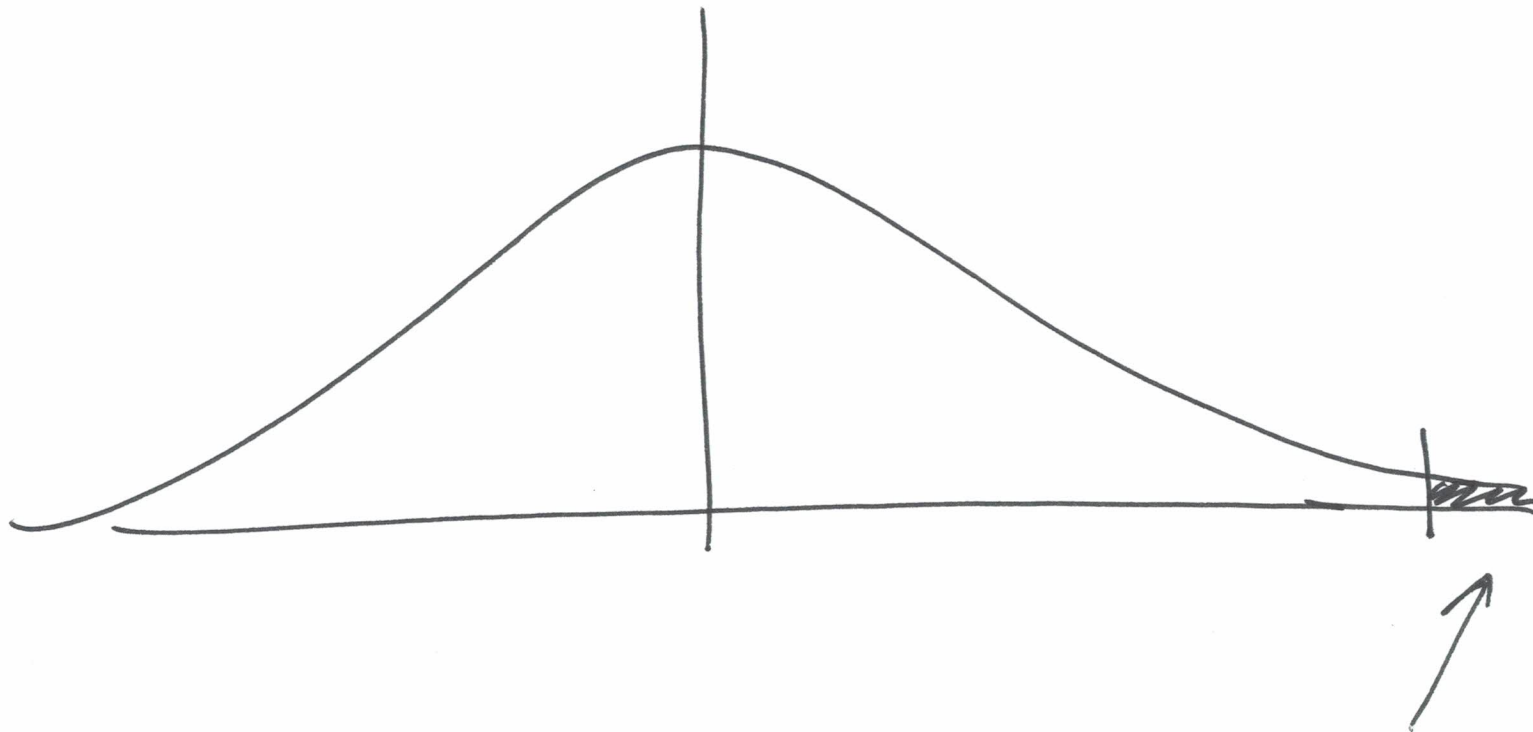
IF WE ACTUALLY KNOW THE PDF,
WE CAN GET A DIRECT ANSWER.

e.g. SAY X IS ACTUALLY UNIFORM ON
 $[228, 372]$. THIS R.V HAS MEAN
300 AND $\sigma^2 = 1728$.



$$= \frac{22}{144} = 0.153$$

BOUND WAS NOT VERY TIGHT!



PREVIEW: WE CAN
ALSO GET ACCURATE
ESTIMATES FROM Q
TABLES FOR CERTAIN
PROBLEMS.

BOUNDS ARE
MOST USEFUL
FOR VERY
LONG-TAILED
EVENTS.

WITHDRAW/DROP DEADLINE MARCH 20.

EXAM 2 THURSDAY MARCH 26
IN CLASS.